

Abdullah Ali

MACHINE LEARNING ENGINEER ·

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Summary

AI Engineer with experience designing and deploying AI Agents & Agentic RAG systems, and intelligent applications using advanced LLM frameworks. Skilled in deep learning with Transformers, CNNs, NLP, and TensorFlow, building scalable production-ready Sol. with Python, FastAPI, and Docker. Proficient in MLOps, MLflow, CI/CD, and AWS cloud infrastructure, with a focus on developing multimodal AI systems that solve real-world problems.

Tools & Technologies

• Languages & Libraries:

Python – SQL – Langchain – LangGraph (Learning) – Tensorflow – Scikit-learn – NumPy – Pandas – Git – Matplotlib – FastAPI – Pydantic – Optuna

• AI & Machine Learning:

Agents – RAG – LLMs – Fine-Tuning – Transformer – NLP – ANNs – Supervised & Unsupervised Learning – Hyperparameter Tuning – Model Evaluation – EDA – Feature Engineering

• MLOps & AWS Cloud:

MLflow – MySQL – VS Code – Apache Airflow – Docker – CI/CD – EC2 – S3 – RDS – DynamoDB – Lambda – ECS – EKS – IAM – CloudWatch

projects

NexusAI — Agentic RAG Assistant | GitHub

- Built an **agentic RAG AI** assistant that autonomously determines user intent, selects appropriate tools, and performs multi-step reasoning to answer questions using uploaded documents, YouTube content, web information, and external data.
- Engineered **hybrid RAG** combining semantic and keyword retrieval for context-grounded answers across multiple documents, YouTube content, and web/blog sources.
- Implemented JWT-based **authentication**, user-isolated data, persistent SQLite memory, and production API services,
- Developed a secure multi-user platform with **persistent** & isolated workspaces; containerized and deployed on AWS with Docker.

Tools: Python, **LangGraph**, **LangChain**, FastAPI, Streamlit, FAISS, **BM25**, Gemini, Ollama, SQLite, Docker, AWS EC2, **JWT**

AI Medical Assistant | GitHub

- Developed a multimodal AI assistant supporting **text, voice, image and Pdfs**, and medical-report inputs, combining LLM reasoning, computer vision, speech processing, and **RAG-based** document understanding
- Engineered a **medical report** RAG pipeline enabling users to upload reports and query their contents through **semantic retrieval** and LLM-powered, context-grounded responses, with modular multimodal processing workflows
- Containerized and deployed application on AWS EC2 using Docker, delivering a cloud, production-ready AI inference service

Tools: Python, LLMs, **Groq**, Computer Vision, Speech-to-Text, Text-to-Speech, **Gradio**, AWS, FastAPI, Streamlit, **SQLite**, Docker

Network Security | Github

- Developed an end-to-end machine learning system to detect phishing websites using 30+ engineered features.
- **Achieved 98% accuracy** with tuned Random Forest among multiple classifiers (Decision Tree, Logistic Regression, Boosting).
- Built **modular pipelines** for ingestion, validation, transformation, and training with full schema and artifact tracking.
- Deployed the solution on AWS EC2 with FastAPI, Streamlit, Docker, GitHub Actions (CI/CD), MLflow, DVC, and S3..

Tools: Python, **FastAPI**, **Scikit-learn**, Streamlit, MLflow, DVC, **AWS** (S3, EC2), Docker, GitHub Actions, **MongoDB Atlas**, Git, Pandas

Certifications

- **Complete Agentic AI Bootcamp** (*Certificate*)
- **Complete MLOps Bootcamp** (*Certificate*)
- **Machine Learning A-Z** (*Certificate*)

Academics

Virtual University of Pakistan

BACHELOR IN COMPUTER SCIENCE

Lahore, Pakistan